CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 23: Graphical Representation (Histograms and Ogives)

Every formula and result of the chapter on one sheet

1	Class mark	$\frac{\text{lower limit} + \text{upper limit}}{2}$
2	Class size	Upper limit − lower limit (continuous classes)
3	Adjustment factor	$\frac{\text{lower limit of a class} - \text{upper limit of the previous class}}{2}$
4	Histogram	Rectangles on continuous classes; height = frequency; no gaps
5	Mode from a histogram	Foot of the perpendicular from the crossing point of the two diagonal lines in the tallest rectangle
6	Ogive	Plot (upper class limit, cumulative frequency); join with a smooth curve
7	Starting point of an ogive	(lower limit of the first class, 0)
8	Median from an ogive	Value at cumulative frequency $\frac{N}{2}$
9	Lower quartile Q_1	Value at cumulative frequency $\frac{N}{4}$
10	Upper quartile Q_3	Value at cumulative frequency $\frac{3N}{4}$
11	Inter-quartile range	$Q_3 - Q_1$
12	Number above a value	N − cumulative frequency at that value